

Computer Engineering Department

King Mongkut's University of Technology Thonburi

CPE 241 Database

Problem Session #10 User Interface Design for Database Systems

Date: Check PS10 assessment activities on LEB2

Introduction

In this lab, you will explore key features of relational database management systems (RDBMS), with a focus on ensuring data integrity, access control, and performance optimization. These features are fundamental to building reliable, secure, and efficient database applications. You will work hands-on with the following concepts:

- Transactions – Learn how to group multiple SQL operations into a single unit of work that can be committed or rolled back to maintain database consistency.
- Users and Privileges – Create user accounts and assign fine-grained access rights to restrict or allow certain database operations.
- Views – Create virtual tables to simplify queries and control access to sensitive data.
- Indexes – Implement indexes to improve the speed of data retrieval for frequently queried columns.
- Query Analysis – Use **EXPLAIN** and **ANALYZE** to understand how DBMS executes queries and determine whether your indexes are being used effectively.

Objectives

1. Manage transactions using **START TRANSACTION**, **COMMIT**, and **ROLLBACK**
2. Control user access using **GRANT** and **REVOKE**
3. Create and use views to restrict data exposure
4. Design and apply indexes for performance enhancement
5. Analyze query execution plans to identify bottlenecks

Instructions

STEP 1: Setup – Create Database and Table

```
CREATE DATABASE company_db;
USE company_db;
CREATE TABLE employee (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100),
    position VARCHAR(50),
    department VARCHAR(50),
    salary DECIMAL(10,2),
    hire_date DATE
);
```

STEP 2: Transactions

```
START TRANSACTION;
INSERT INTO employee (name, position, department, salary,
hire_date)
VALUES ('Alice', 'Manager', 'HR', 95000, '2020-01-15');
INSERT INTO employee (name, position, department, salary,
hire_date)
VALUES ('Bob', 'Engineer', 'IT', 105000, '2021-03-01');
ROLLBACK;
```

```
-- Now with correct data
START TRANSACTION;
INSERT INTO employee (name, position, department, salary,
hire_date)
VALUES ('Bob', 'Engineer', 'IT', 98000, '2021-03-01');
COMMIT;
```

STEP 3: User and Privileges

```
CREATE USER 'hr_user'@'localhost' IDENTIFIED BY 'hrpass123';
GRANT SELECT, INSERT ON company_db.employee TO
'hr_user'@'localhost';
FLUSH PRIVILEGES;
```

STEP 4: Create View

```
CREATE VIEW employee_public AS
SELECT id, name, position, department FROM employee;
GRANT SELECT ON company_db.employee_public TO
'hr_user'@'localhost';
```

STEP 5: Add Indexes

```
-- Index on department for faster filtering
CREATE INDEX idx_department ON employee(department);

-- Composite index on position and hire_date
CREATE INDEX idx_pos_date ON employee(position, hire_date);
```

STEP 6: Analyze Queries with EXPLAIN and ANALYZE

```
-- Use EXPLAIN to see the query plan
EXPLAIN SELECT * FROM employee WHERE department = 'IT';
-- Use ANALYZE (in MySQL 8.0+) to measure actual execution
ANALYZE FORMAT=TRADITIONAL
SELECT * FROM employee WHERE department = 'IT';
```

Deliverables

Students must submit a report detailing:

1. Step-by-Step Screenshots

For each lab section, include screenshots of your SQL commands and their results in the DBMS terminal (or GUI client like MySQL Workbench). The required screenshots are

- Transaction commands:
 - **START TRANSACTION**, **COMMIT**, and **ROLLBACK**
 - Show the effect of a rollback (data not saved)
 - Show the effect of a commit (data saved)
- User and Privilege Management
 - Creating a new user
 - Granting and revoking privileges
 - Testing access from the new user account (e.g., denied query)

- View Usage
 - Creating the view
 - Querying the view as a limited-access user
 - Attempting to access the main table (and getting permission error, if possible)
- Index Creation
 - Creating the view
 - SQL command used to create index
 - `SHOW INDEX FROM employee;` output to verify index
- Query Analysis
 - Output from `EXPLAIN` or `ANALYZE` for: a query that uses an index and a query that does not use an index (for comparison)

2. Reflection for Each Section

After each screenshot or group of screenshots, write a brief reflection (2–5 sentences) describing:

- What you did in this step
 - What you learned or observed (e.g., how rollback canceled changes, how index improved query speed)
 - Any challenges you faced or interesting outcomes
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Final Submission

- Compile all written explanations and/or pictures into a single a PDF file and ensure it is named appropriately (e.g., `PS10_YourStudentID.pdf`) then submit the report on LEB2 before the deadline.
- Upload the PDF report to the Assessment Activities section for PS10 on LEB2.